

Effect of potassium promoter on precipitated iron-manganese catalyst for Fischer–Tropsch synthesis

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Abstract

A systematic study has been carried out to investigate the impact of potassium promoter on the performance of a precipitated iron-manganese catalyst for Fischer–Tropsch synthesis (FTS). Characterization technologies of N_2 physisorption, X-ray diffraction (XRD), Mössbauer effect spectroscopy (MES) and H_2 thermal gravimetric analysis (H_2 -TGA) were used to study the effect of potassium on the textural properties, bulk phase composition and reduction behavior. FTS reaction test was performed in a fixed bed reactor. The results of characterization showed that the addition of potassium leads to the relatively large crystallite size of α - Fe_2O_3 and inhibits the reduction of catalyst. The carbonization of the catalyst is enhanced with the increase in both the potassium content and the reaction temperature. A maximum in FTS and water-gas shift (WGS) activity is noted upon increasing K content (0.7 wt.% K), followed by a sharp decline in activity at the potassium level in excess of the maximum. It is found that potassium is an effective promoter to suppress the hydrogenation function of the catalyst. The selectivity to olefins is promoted and the formation of methane and light hydrocarbons is restrained with the increasing potassium level. The selectivity to oxygenates shows a rapid and monotonic decrease with the increase of potassium loading and passes through a minimum at potassium loading of 0.7 wt.%. After the point, it increases slowly with further increasing in potassium content. At the same time, increasing reaction temperature results in a monotonic decrease in the weight percent of oxygenates over the un-promoted and potassium-promoted catalysts. © 2004 Elsevier B.V. All rights reserved.

Keywords: Fischer–Tropsch synthesis; Potassium promoter; Iron-manganese catalyst; Mössbauer effect spectroscopy; Hydrocarbon distribution

1. Introduction

Fischer–Tropsch synthesis for converting syngas to hydrocarbons has been attracted much interest as an important process for the production of transportation fuels and chemicals. Iron-based FTS catalysts provide both hydrogenation and water-gas-shift activities, imposing a flexible option as a working catalyst for typically coal-derived CO-rich syngas conversion [1,2]. Iron-based catalysts often contain small amounts of potassium and some other metals such as manganese, calcium, zinc, copper and magnesium as promoters to improve its activity and selectivity [3], since potassium has the stronger basicity and influences the adsorption of reactants (CO and H_2) on the active sites. This leads to some effects on the FTS activity, the enhancement in the selectivity to olefins, the suppression of the formation of methane,

and the selectivity shift to higher molecular weight products [4–6]. Nearly all iron-based FTS catalysts contain more or less potassium as one of the promoters. The overall effects of potassium on the behavior of iron-based FTS catalysts have been extensively investigated and are well established over different catalyst systems [3–19]. Kölbel [4] investigated the effect of potassium on the surface properties over supported iron and precipitated Fe–Cu– SiO_2 catalysts, and found that the addition of potassium on the precipitated iron catalyst enhanced CO chemisorption and suppressed H_2 chemisorption. The results were explained by the fact that potassium donates electrons to iron and facilitates CO chemisorption, since CO tends to accept electrons from iron. On the other hand, hydrogen with higher surface coverage was inclined to donate electrons to iron, and the electrons donated to iron from potassium weakened the strength of the Fe–H bond. Hence, potassium strengthened the Fe–C bond and weakened the Fe–H bond [9,11,14,20]. In a more recent study, Jiang et al. [17] investigated the effects of potassium on iron-manganese catalysts by in situ diffuse reflectance

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FT-IR using NO, CO and CO + H₂ as probes. The results of Jiang indicated that the potassium species tightly interact with the surface iron species and promote iron oxide reduction to form the fine metallic iron clusters. No effect of potassium on FTS performance was mentioned in the paper; however, XPS measurements on the K 2p levels showed that the potassium associates in some form with oxygen. It is not covered by the deposited carbon, but appears on the top of the carbon layer [18]. At the same time, the results of Rankin and Bartholomew over an Fe/SiO₂ catalyst indicated that the addition of potassium restrains the reduction of catalyst [19]. Dry and Oosthuizen [8] have reported that the addition of potassium results in a decrease in the surface area of fused magnetite catalysts. In addition, the carbonization of iron during FTS proceeded more rapidly on potassium-promoted catalysts [6]. The effects of potassium on the FTS and WGS activities and product selectivity have also been investigated over a variety of iron-based catalysts [14,18,20,21]. These could be summarized as follows: (1) improving the dissociative adsorption of CO, strengthening the bond of Fe–C, and facilitating carbon deposition and catalysts deactivation; (2) increasing the selectivity of olefins, suppressing the formation of methane, and shifting to the higher molecular products; (3) increasing the activity of WGS; and (4) influencing FTS activity. The FTS activity either increases [3,7,14] or passes through a maximum as a function of potassium loading [12,21], and potassium either has no effect on the activity for FTS [22] or suppresses it [18,21]. Manganese has been widely used as one of the promoters for FTS on iron catalysts, particularly in producing low olefins [23–26]. Large efforts have also been made on the individual effect of manganese promotion on supported or unsupported iron catalysts [26,27]. Although the effects of potassium promotion on the FTS performances of iron-based catalysts have been extensively investigated in the above studies, limited results have been reported for the effects of potassium on the manganese-promoted precipitated iron catalysts [13,28].

The present work focuses on a systematic understanding of the effects of potassium promotion on precipitated iron-manganese catalysts under industrial relevant operation conditions. Particular attention is given to the effects of potassium on the catalyst reduction, the textural properties and the bulk phase structure and to the compositions of precipitated Fe/Mn catalyst as prepared, after reduction and after FTS reaction. The FTS and WGS activity, olefin and oxygenate selectivities, and hydrocarbon product distribution are correlated with catalyst characterization results.

2. Experimental

2.1. Catalyst preparation

The Fe/Mn/K catalysts used in the present study were prepared by the co-precipitation of Fe and Mn nitrates at a constant pH value to form the precursor of Fe-Mn oxyhy-

droxide, which was promoted by impregnation with K₂CO₃ solution after thermal treatment in air. A solution containing both Fe(NO₃)₃ and Mn(NO₃)₂ and a separate solution of NH₄OH were used in the precipitation processes. The salt solution and ammonia solution were preheated to 353 and 313 K, respectively. Under stirring, the Fe/Mn solution with a certain flow rate was added into a continuously stirred tank reactor containing 200 cm³ ammonium hydroxide solution with pH 8.8–9.0. Additional amount of the ammonium solution was added simultaneously, with a flow rate to keep the pH within 8.8–9.0. The temperature of the precipitation unit was kept at 351–353 K during the whole precipitation process. The precipitate was filtered, washed completely with de-ionized water, and dried in air at 393 K for 36 h. The potassium promoter was added in a calculated amount to the precursor by impregnation with aqueous solution of K₂CO₃ to give the desired K content. The weight percents of potassium for five catalyst samples are designed to be 0, 0.2, 0.7, 1.5 and 3.0 wt.%, respectively. The excess solvent in the impregnated samples was removed by evaporating at 353 K, and the final products were further dried in air at 393 K for 24 h and then calcinated at 773 K for 5 h. All samples were pressed into pellets (60 MPa), crushed and sieved to retain 20–40 mesh particles for reaction tests.

2.2. Catalyst characterizations

BET surface area, pore volume and average pore diameter of fresh catalysts were measured by the BET method using a Micromeritics ASAP 2500 for N₂ physical adsorption at its normal boiling point (77 K) after the samples were degassed at 393 K for 6 h. The surface atomic ratios of the fresh catalysts were determined by XPS using a PHI-5300/ESCA spectrometer with Al K α radiation; the spectrometer resolution of energy was 0.8 eV. The peak positions were corrected for sample charging by setting the C 1s binding energy at 284.6 eV. The relative surface concentrations of the metals were determined using the whole peak area of Fe 2p, Mn 2p and K 2p regions and the corresponding sensitivity factors of Wager et al. [29].

The thermogravimetric analysis (TGA) was performed using an MS OmniStar 200 instrument. Typically, 20–30 mg samples were treated in 50% H₂ (by mole basis) mixed with Ar at ambient conditions for 10 min and then temperature was increased from room temperature to 1073 K at a rate of 10 K/min and held for 5 min before cooling.

Powder X-ray diffraction (XRD) measurements were carried out using a D/max-RA X-ray diffractometer (Rigaku, Japan) with Cu K α radiation (λ = 0.154 nm) operated at 40 kV and 100 mA. The Mössbauer spectra of the catalysts were obtained at room temperature by using an Austin-S-600 constant-acceleration Mössbauer spectrometer (Austin, USA). The γ -ray source was ⁵⁷Co/Pd. All spectra were analyzed with a set of independent Lorentzian lines and with the help of a non-linear least squares fitting procedure. The number of center channels and the increment

of velocity per channel were determined by a standard α -Fe sample at room temperature. Isomer shifts were reported with respect to α -Fe sample (99.99%, Alfa Aesar). Usually it was assumed that the Mössbauer area ratios are equal to a relative amount of the associated species.

The reduced catalyst samples used for XRD and MES characterization were prepared by reducing the fresh catalysts in a quartz tube with synthesis gas ($H_2/CO = 2$) at 523 K and 0.1 MPa for 32 h. After reduction, the reactor was sealed in the inert atmosphere and then transferred to a glove box. The samples were transferred to glass tubes and coated with paraffin wax for preventing the oxidation under the Ar atmosphere and then sealed for characterization.

2.3. Reactor system and operation procedure

A detailed description of the reactor and the product collection system was given by Ji et al. [30]. Briefly, experiments were conducted in a 12-mm i.d. stainless steel fixed bed reactor with an effective bed length of approximately 15 cm (15 cm^3 bed volume), which was placed within a furnace equipped with a temperature controller. The bed temperature was monitored axially using a K-type thermocouple. The feed gas with a H_2/CO ratio of approximately 2.0 prepared by decomposing methanol passed through a series of columns, an activated charcoal trap, an oxygen-removal trap, a sulfur-removal trap and a silica-gel/5A molecular sieve trap, to remove tiny amounts of carbonyls, oxygen, sulfur and water before it enters the reactor. The flow rate of the purified syngas was controlled by using a mass flow controller. The outlet of the reactor is connected with a hot trap (393 K) and a cold trap (273 K) at the system pressure. After the product collectors, the pressure of the tail-gas was released through a backpressure regulator. The flow rate of tail gas was monitored by a wet-gas flow meter.

Typically, for all reaction experiments, 5 ml catalyst was charged in the reactor. The remaining volume of the reactor tube was filled with ceramics beads in a diameter range of 10–20 mesh. Before reaction, all catalysts were reduced with syngas ($H_2/CO = 2.0$) at 523 K, 0.10 MPa and 1000 h^{-1} for 32 h. Following activation, the bed was cooled to 473 K. The system was then pressurized to 2.50 MPa. The temperature was gradually increased to 523 K. Following the change of reaction conditions, a 12–16 h was needed to attain the steady state of the system. After this unsteady state period, the products in the hot and cold traps are collected over 24 h (mass balance period), weighted, and sampled for analysis. During the mass balance period, the tail gas was analyzed twice.

The product of FTS included gas phase, liquid phase (in the ice trap) and wax phase (in the hot trap). Both the purified feed gas and the tail gas were analyzed on a gas chromatograph (GC, GC 920, Shanghai Analyzer Co., China). H_2 , CO, CH_4 , and tiny of N_2 and O_2 were separated on a GC 920 with a 13X molecular sieve packed column ($1.5\text{ m} \times 3\text{ mm}$ i.d., Ar carrier) attached to a thermal conductivity detec-

tor (TCD). C_1 – C_6 hydrocarbons (*n*-paraffins, *n*-olefins, and branched isomers) in the tail gas phase were analyzed on a Shimadzu-7A GC equipped with a C_{22}^0/C -22 packed column ($7\text{ m} \times 3.2\text{ mm}$ i.d., N_2 carrier) and a flame ionization detector (FID). The amount of CO_2 is measured on a GC 920 with a packed column ($401, 1\text{ m} \times 3\text{ mm}$ i.d., H_2 carrier) and TCD, and quantified by an external standard method. A Shimadzu 17A GC (Shimadzu, Japan) equipped with an FID ($60\text{ m} \times 0.25\text{ mm}$ i.d. OV-101 capillary column under temperature programming, N_2 carrier) was used to analyze the oil phase collected in the cold trap. The analysis of oxygenates in water phase was performed on an SP-520 (Shanghai Analyzer Co., China) gas chromatograph and an FID. High molecular weight products (wax) collected in the hot trap was analyzed on an Angilent 6890N (Angilent, HP) gas chromatograph using an OV-101 capillary column ($15\text{ m} \times 0.53\text{ mm}$ i.d., N_2 carrier) and an FID under temperature programming. The mass of components analyzed was calculated from the known response factors and a known amount of reference compound (*n*- C_{22} paraffin).

To investigate the effect of potassium on the crystallite structures in the used catalysts under the same reaction condition, we performed another set of FTS experiments. The operation process was the same as that mentioned above; the reaction terminated after the mass balance at 553 K, then the samples were cooled to room temperature, and finally the catalysts were drawn from the reactor and characterized by XRD and MES.

3. Results and discussions

3.1. Textural properties and reduction of the catalysts

The BET surface area, pore volume, average pore diameter, Fe/Mn and Fe/K atomic ratios in the bulk and the surface of the catalysts with different potassium levels as prepared are illustrated in Table 1 and Fig. 1. It is apparent that potassium clearly influences the surface area, pore volume and pore size distribution. For more potassium

Table 1
Effect of potassium promotion on the BET surface area, pore volume and average pore diameter of the potassium promoted Fe/Mn catalysts as-prepared

The content of K (wt.%)	0	0.2	0.7	1.5	3.0
Specific area (m^2/g)	52	30	26	19	17
Volume of pore (cm^3/g)	0.27	0.23	0.23	0.18	0.16
Average pore diameter (nm)	20.57	30.85	35.75	34.61	35.86
Fe/Mn atomic ratio					
Bulk	9.0	9.0	9.0	9.0	9.0
Surface	3.4	2.3	2.4	2.0	2.1
K/Fe ($\times 10^2$ atomic ratio)					
Bulk	0	0.5	1.6	3.4	6.9
Surface	0	0.7	2.5	4.7	8.7

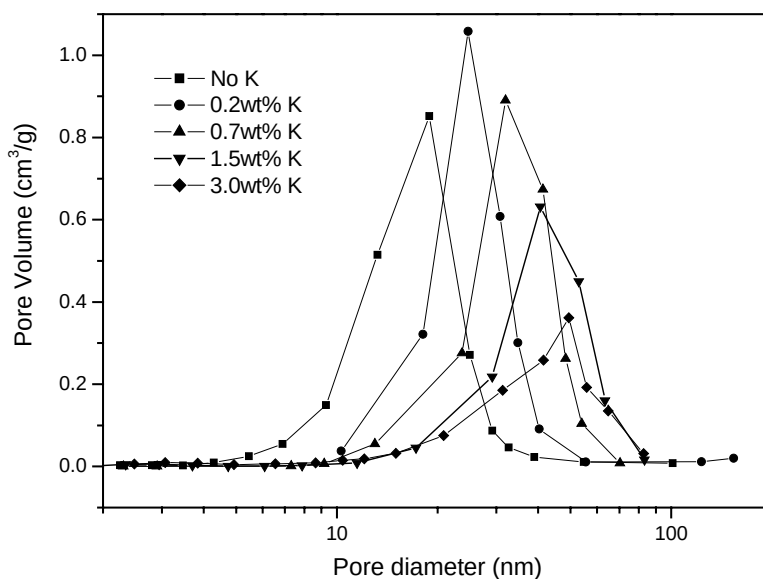


Fig. 1. Pore size distribution of catalysts as prepared with different potassium levels.

loading, the catalysts have the larger pore diameters and lower specific surface areas and pore volumes. The results of Dry over a fused iron catalyst have also led to a similar conclusion: that the more alkali was added, the greater loss in surface area was found [8]. Such a result can be ascribed to the fact that potassium could improve the agglomeration of the FeOOH precursor and could further enlarge the crystallite size of α -Fe₂O₃ after being calcined at 773 K for 5 h, which would induce the decrease in surface area. Li et al. have drawn the opposite result over precipitated Fe/Cu catalyst [31]. This discrepancy may be caused by the different catalyst preparation procedures. They added potassium after the thermal treatment at 623 K for 1 h, and the FeOOH may have been transformed to the stable α -Fe₂O₃ phase. XRD and MES studies in the present study also show that the samples containing potassium have larger crystallites than the sample without potassium. The pore diameter of the catalyst simultaneously increases with the increase of the potassium content. The loss of surface area is hence due to crystallite growth and not due to other reasons such as pore blocking with potassium addition. It is also found that the K/Fe atomic ratio on surface is much larger than in the bulk of samples from the analysis of XPS, which implies that the potassium promoter largely concentrates on the catalyst surface. The Fe/Mn atomic ratio in Table 1 indicates that the addition of potassium may generally induce the enrichment of Mn on the catalyst surface. This has been suspected to be one of the reasons for the improved olefin selectivity over Mn-promoted iron catalyst [25,26].

Fig. 2 shows the effect of potassium on the curves of weight loss vs. temperature in H₂/Ar atmosphere. The three negative peaks observed in DTG pattern for all catalysts represent the reduction steps of α -Fe₂O₃ to Fe₃O₄, Fe₃O₄ to FeO, and FeO to α -Fe, respectively. The net weight loss at the same temperature point decreases with the increase

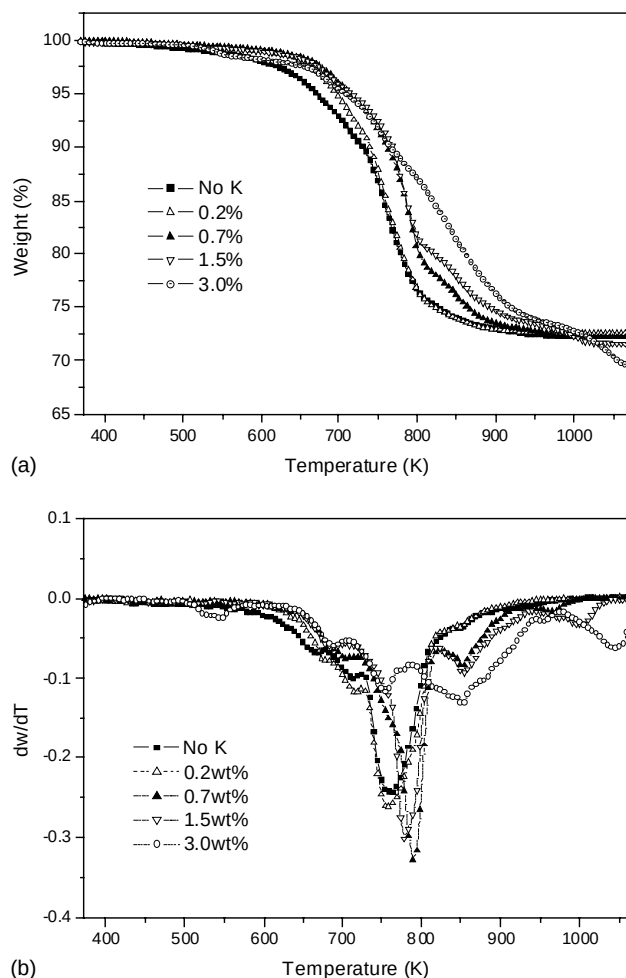


Fig. 2. Thermal gravimetric (TG) and differential thermal analysis (DTA) curves for the catalysts with different potassium levels reduced with 50% H₂ mixed with Ar (by mole basis).

Table 2
Mössbauer parameters of the potassium promoted Fe/Mn catalysts as-prepared

K content in catalysts (%)	Mössbauer parameter			Assignment	Spectral contribution (%)
	IS (mm/s)	QS (mm/s)	Hhf (kOe)		
0	0.53	0.01	501	α -Fe ₂ O ₃ (m)	75.1
	0.27	0.75		α -Fe ₂ O ₃ (s)	24.9
0.2	0.53	0.01	510	α -Fe ₂ O ₃ (m)	93.3
	0.39	0.64		α -Fe ₂ O ₃ (s)	6.7
0.7	0.53	0.01	515	α -Fe ₂ O ₃ (m)	94.0
	0.35	0.70		α -Fe ₂ O ₃ (s)	6.0
1.5	0.53	0.00	516	α -Fe ₂ O ₃ (m)	94.8
	0.32	0.68		α -Fe ₂ O ₃ (s)	5.2
3.0	0.53	0.01	516	α -Fe ₂ O ₃ (m)	95.0
	0.32	0.70		α -Fe ₂ O ₃ (s)	5.0

of potassium levels, indicating that the reduction of catalyst is retarded by the addition of potassium. This may be the result of the strong interaction between potassium oxide and iron oxide. The strong interaction of potassium oxide with iron oxide could suppress the adsorption of hydrogen on catalyst surface and therefore restrain the reduction of iron oxide underlying and/or neighboring the potassium promoter [19,32]. In addition, the increase in α -Fe₂O₃ crystallite size leads to the decrease in the surface area that can contact with H₂ reductant; this could also contribute to the reduction phenomenon observed in Fig. 2. Temperature programmed and isothermal reduction studies at 573 K conducted by Li also revealed that potassium inhibits the

reduction of iron catalyst when H₂ is used as the reductant [33].

3.2. Crystallite structure of the catalysts

Bulk iron phases in catalysts as prepared, after reduction, and after FTS reaction were detected by XRD and Mössbauer effect spectroscopy (MES). Results of these measurements are summarized in Tables 2–4. The XRD patterns and MES spectra are shown in Figs. 3–5 and Figs. 6–8, respectively. The Mössbauer spectra display doublets, sextets and their combinations. The X-ray diffraction spectra have been plotted over 2θ values ranging from 20 to 75°.

Table 3
Mössbauer parameters of potassium promoted Fe/Mn catalysts after reduction

K content in catalysts (%)	Mössbauer parameter			Assignment	Spectral contribution (%)
	IS (mm/s)	QS (mm/s)	Hhf (kOe)		
0	0.43	0.06	497	Fe ₃ O ₄ (A)	52.6
	0.65	0.09	455	Fe ₃ O ₄ (B)	30.9
	0.28	0.23	190	II in χ -Fe ₅ C ₂	2.5
	0.34	0.67		Fe ³⁺ (s)	14.1
0.2	0.44	0.03	500	Fe ₃ O ₄ (A)	62.6
	0.66	0.09	454	Fe ₃ O ₄ (B)	34.1
	0.28	0.23	191	II in χ -Fe ₅ C ₂	0.9
	0.34	0.67		Fe ³⁺ (s)	2.3
0.7	0.43	−0.05	507	Fe ₃ O ₄ (A)	63.6
	0.65	0.13	461	Fe ₃ O ₄ (B)	32.3
	0.27	0.23	189	II in χ -Fe ₅ C ₂	1.2
	0.35	0.63		Fe ³⁺ (s)	2.9
1.5	0.46	−0.06	510	Fe ₃ O ₄ (A)	67.7
	0.63	0.02	464	Fe ₃ O ₄ (B)	29.6
	0.29	0.25	190	II in χ -Fe ₅ C ₂	0.5
	0.38	0.59		Fe ³⁺ (s)	2.2
3.0	0.46	−0.07	512	Fe ₃ O ₄ (A)	69.2
	0.66	0.09	465	Fe ₃ O ₄ (B)	27.1
	0.27	0.22	188	II in χ -Fe ₅ C ₂	0.8
	0.34	0.69		Fe ³⁺ (s)	2.9

Reduction condition: 523 K, 0.1 MPa, H₂/CO = 2.0, 1000 h^{−1} and 32 h.

Table 4

Mössbauer parameters of the co-precipitated Fe/Mn catalysts after reaction

K content in catalysts (%)	Mössbauer parameters			Assignment	Spectral contribution (%)
	IS (mm/s)	QS (mm/s)	Hhf (kOe)		
No K (553 K)	0.30	0.01	491	Fe ₃ O ₄ (A)	43.8
	0.62	0.04	456	Fe ₃ O ₄ (B)	54.2
	0.23	0.52	209	I in χ -Fe ₅ C ₂	0.6
	0.28	0.23	176	ϵ' -Fe _{2.2} C	1.4
0.2 wt.% (553 K)	0.29	0.03	491	Fe ₃ O ₄ (A)	38.1
	0.64	0.03	456	Fe ₃ O ₄ (B)	60.1
	0.32	0.35	208	I in χ -Fe ₅ C ₂	1.7
	0.27	0.45	173	ϵ' -Fe _{2.2} C	0.2
0.7 wt.% (553 K)	0.30	0.02	493	Fe ₃ O ₄ (A)	35.6
	0.66	0.06	462	Fe ₃ O ₄ (B)	44.5
	0.32	0.35	208	I in χ -Fe ₅ C ₂	14.7
	0.27	0.20	173	ϵ' -Fe _{2.2} C	5.2
1.5 wt.% (553 K)	0.41	0.05	474	Fe ₃ O ₄ (A)	37.9
	0.78	0.05	441	Fe ₃ O ₄ (B)	43.1
	0.32	0.35	208	I in χ -Fe ₅ C ₂	15.0
	0.27	0.20	173	ϵ' -Fe _{2.2} C	4.0
3.0 wt.% (553 K)	0.31	0.03	493	Fe ₃ O ₄ (A)	35.9
	0.67	0.00	459	Fe ₃ O ₄ (B)	32.6
	0.31	0.11	223	I in χ -Fe ₅ C ₂	22.1
	0.30	0.20	176	ϵ' -Fe _{2.2} C	9.5
0.2 wt.% (573 K)	0.31	0.04	493	Fe ₃ O ₄ (A)	49.5
	0.66	0.01	457	Fe ₃ O ₄ (B)	43.3
	0.29	0.14	220	I in χ -Fe ₅ C ₂	5.6
	0.3	0.30	176	ϵ' -Fe _{2.2} C	1.7
1.5 wt.% (593 K)	0.43	0.12	495	Fe ₃ O ₄ (A)	38.7
	0.73	0.20	456	Fe ₃ O ₄ (B)	28.4
	0.5	0.17	211	I in χ -Fe ₅ C ₂	16.5
	0.43	0.21	163	ϵ' -Fe _{2.2} C	16.4

Reaction condition: 2.50 MPa, H₂/CO = 2.0, 1000 h⁻¹.

3.2.1. Catalysts as prepared

The powder X-ray diffraction patterns for the Fe/Mn catalysts with potassium content of 0–3.0 wt.% are shown in

Fig. 3. The only detectable phase identified in the diffraction patterns of the catalysts with different potassium contents as prepared is hematite (α -Fe₂O₃), which has characteristic

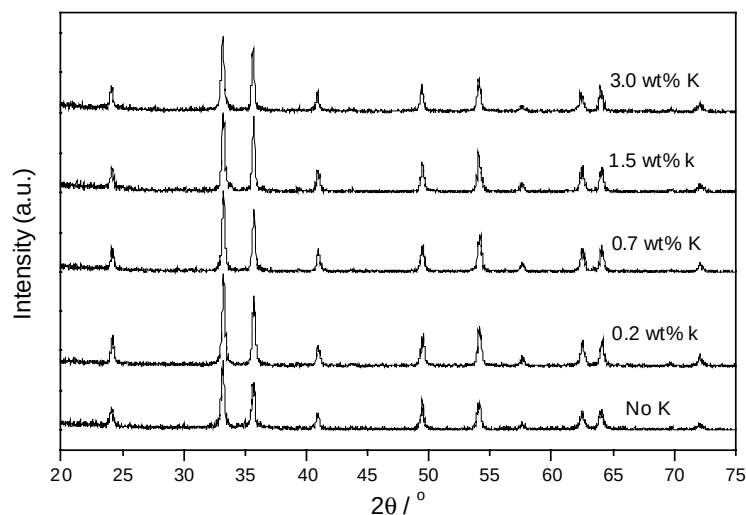


Fig. 3. X-ray diffraction patterns of the potassium promoted Fe/Mn catalysts as-prepared.

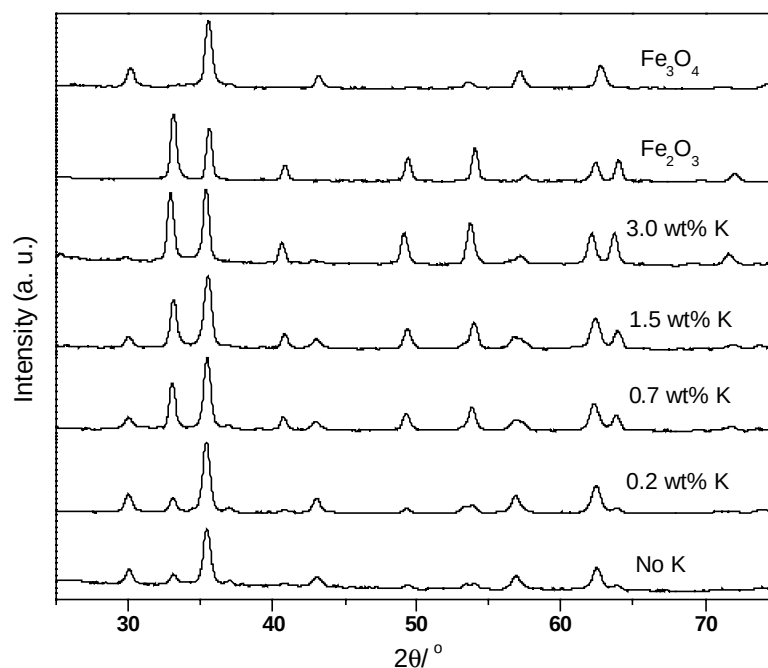


Fig. 4. X-ray diffraction patterns of the potassium promoted Fe/Mn catalysts after reduction.

peaks at 2θ values of 24.2, 33.1, 35.6, 40.8, 49.52, 54.0, 57.6, 62.5 and 64.0°. The peak intensity of α -Fe₂O₃ in samples promoted with potassium is larger than that of the un-promoted catalyst. It implies that the addition of potassium into the iron-manganese catalyst promotes the aggregation of α -Fe₂O₃ crystallite, and therefore the diffraction peak intensity of α -Fe₂O₃ increases with the addition of

potassium. This is consistent with the observed decrease in the catalyst surface area. The MES spectra of the catalysts as prepared shown in Fig. 6 include a sextet and a doublet. According to the MES parameters listed in Table 2, the sextet is assigned to the magnetic α -Fe₂O₃ of large crystallites. The doublet is typical for the superparamagnetic Fe³⁺ ions on the non-cubic sites [34] with the crystallite diameters smaller

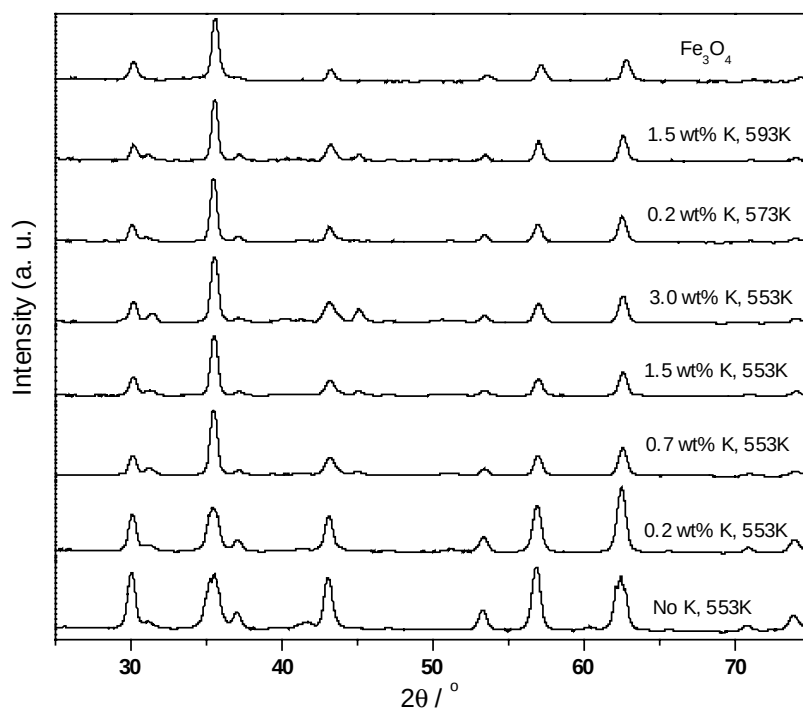


Fig. 5. X-ray diffraction patterns of the potassium promoted Fe/Mn catalysts after reaction.

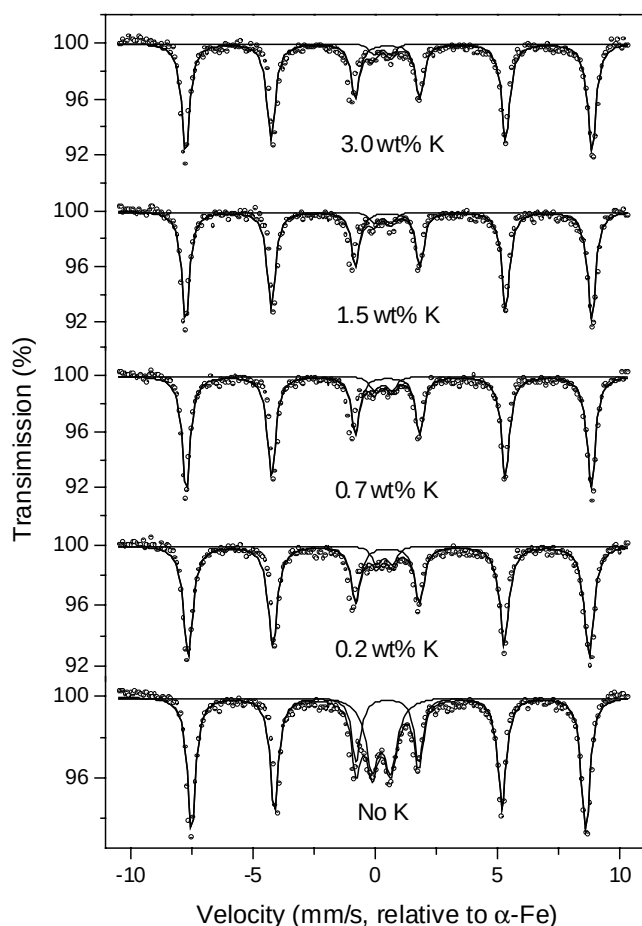


Fig. 6. Mössbauer spectra of the potassium promoted Fe/Mn catalysts as-prepared.

than 13.5 nm [16]. This identification is confirmed by XRD patterns. According to MES spectral area, the un-promoted catalyst has 75.1% ferromagnetic Fe^{3+} and 24.9% superparamagnetic Fe^{3+} . With the increase in the potassium content, the percentage of superparamagnetic Fe^{3+} decreases, while that of ferromagnetic Fe^{3+} increases and attains to 95% at the largest potassium loading of 3.0 wt.%. Furthermore, the hyperfine field (Hhf) of sextet increases with the increase in the potassium loading, and comes up to the standard value of pure magnetic $\alpha\text{-Fe}_2\text{O}_3$ of 515 kOe [35]. It is known that the hyperfine field of $\alpha\text{-Fe}_2\text{O}_3$ increases with the increase of crystallite size due to superparamagnetic effects. At the same time, the value of the quadrupole splitting (QS) of sextet is close to zero, which implies that the iron atoms are located at cubic symmetrical sites [35]. The results implied that the addition of potassium to the catalyst results in the magnetic structure change from superparamagnetic to ferromagnetic due to the increase in crystallite size [16]. The results of MES agree well with those of XRD and BET.

3.2.2. Catalysts after reduction

The XRD and MES patterns of the catalysts with different potassium contents after reduction with syngas ($\text{H}_2/\text{CO} =$

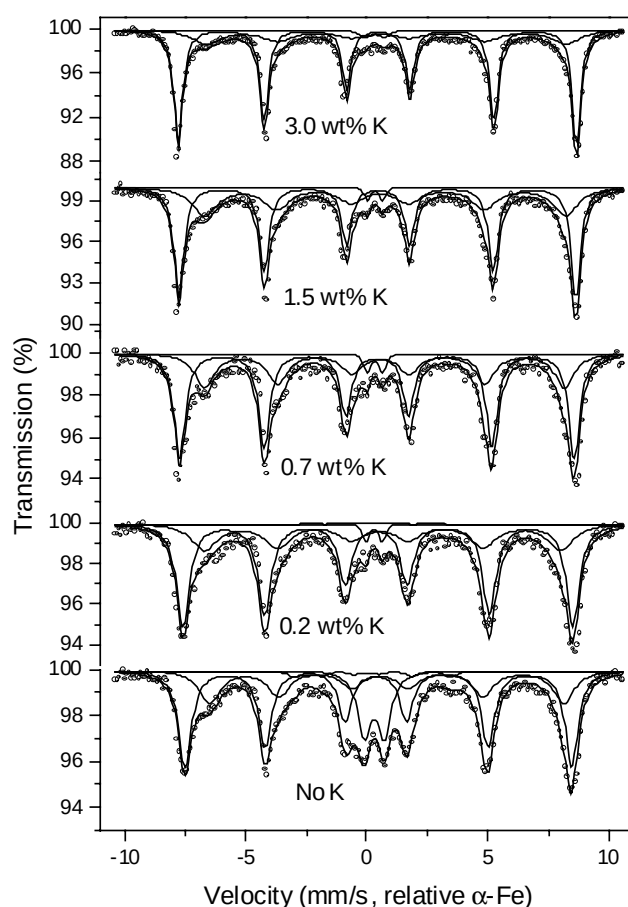


Fig. 7. Mössbauer spectra of the potassium promoted Fe/Mn catalysts after reduction.

2.0) at 523 K, 0.1 MPa and 1000 h^{-1} for 32 h are shown in Figs. 4 and 7, respectively. The MES parameters are summarized in Table 3. The XRD patterns of the reduced catalysts indicate that the reduced catalysts are mainly composed of magnetite (Fe_3O_4) and hematite ($\alpha\text{-Fe}_2\text{O}_3$). Due to the poor crystallographic form of iron carbides and probably very limited amount of iron carbides formed during reduction, there are no iron carbides that can be detected, which could subsequently be confirmed by MES measurement. The XRD patterns also indicate that the peak intensity of $\alpha\text{-Fe}_2\text{O}_3$ clearly increases and that of Fe_3O_4 decreases with the increasing of potassium content. Such results demonstrate that the addition of potassium restrains the reduction of catalysts.

The Mössbauer spectra (Fig. 7) of the reduced samples can be fitted adequately by three sextets and a central doublet; the spectral parameters are summarized in Table 3. The sextets with Hhf of 497–512 and 454–465 kOe can be attributed to the tetrahedral (A site) and octahedral sites (B site) of Fe_3O_4 , respectively. The former represents the Fe^{3+} at the tetrahedral site (A site), and the latter the Fe^{3+} and Fe^{2+} at octahedral site (B site) in Fe_3O_4 [36]. The spectral parameters of the central doublet are close to those of the doublet in the spectra of catalyst samples as prepared and can

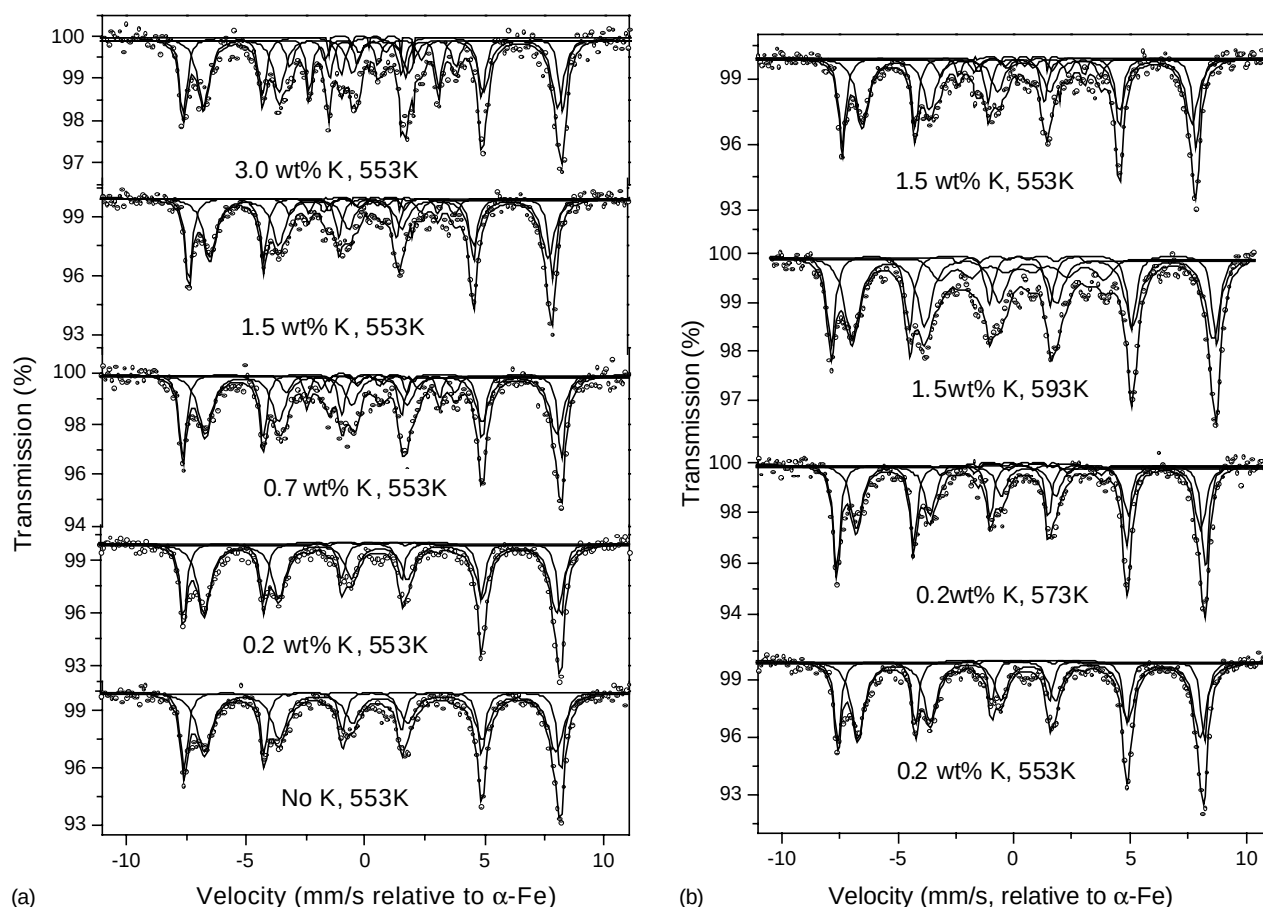


Fig. 8. Mössbauer spectra of the potassium promoted Fe/Mn catalysts after reaction. (a) After reaction at 553 K, (b) MES spectra of 0.2 wt.% K promoted catalysts after reaction at 553 and 573 K, and of 1.5 wt.% K promoted catalysts after reaction at 553 and 593 K, respectively.

be attributed to the small particle octahedral Fe^{3+} ion of superparamagnetic state in small crystallites. The values of the sextet with isomer shift of 0.23–0.25 mm/s, quadrupole splitting of 0.27–0.28 mm/s and hyperfine field of 188–191 kOe imply the presence of $\chi\text{-Fe}_5\text{C}_2$ [23,37,38]. The content of iron carbide is small in each catalyst after reduction, and there is no apparent difference in the carbonization extent under the present reduction conditions. Comparing the areas of the double sextets ascribed to Fe_3O_4 , one finds that the ratio of octahedral site (B site) area to tetrahedral site (A site) area is far lower than the theoretical value of 2.0. The ratio of area B (S_B) to area A (S_A) decreases from 0.59 to 0.39 with the increase of potassium loading. It implies that the catalysts are far away from the complete reduction to magnetite under the above-mentioned reduction conditions. The extent of reduction decreases with the addition of potassium. The result is in agreement with those derived from TG and X-ray diffraction studies mentioned above. The phenomenon has also been reported by Rankin and Bartholomew [19]. As mentioned above, this effect may be a result of a strong interaction between potassium oxide and unreduced iron oxide. Such strong interaction could inhibit the reduction of the iron underlying or adjacent to the promoter [19,32].

3.2.3. Catalysts after reaction

The XRD and MES patterns of the catalysts after reaction are shown in Figs. 5 and 8, and the MES parameters are presented in Table 4. There are two broad and weak peaks at 2θ of 31 and 44.5° along with the characteristic peaks of Fe_3O_4 in the XRD patterns. Since most of the carbide phases reported in the JCPDS [37] database have prominent peaks at 31 and 44.5° , the broad peaks around 31 and 44.5° may be assigned to the presence of iron carbides (perhaps $\chi\text{-Fe}_5\text{C}_2$ and $\epsilon'\text{-Fe}_{2.2}\text{C}$). Due to the poor crystallographic form of iron carbides, the peaks around 31 and 44.5° are broad, and it is impossible to identify which carbide is present in the XRD patterns, or to determine the stoichiometry of those carbides from the XRD patterns. However, the stoichiometry of iron carbides can be determined subsequently by MES parameters. Nevertheless, the XRD patterns also indicate that the diffraction peak intensities of iron carbides are enlarged with the increasing potassium loading under the same reaction conditions (2.50 MPa, 1000 h^{-1} , and 553 K). By comparing the XRD patterns of 0.2 and 1.5 wt.% K-promoted catalysts after different reaction temperatures, it is found that there is no obvious difference in the intensity of Fe_3O_4 and carbides diffraction peaks for the two catalysts. However, with the aid of

succeeding MES analysis, the effect of reaction temperature on the content of carbides is obvious. The iron carbides content are 1.9 and 19.0% after the reaction at 553 K, and increase to 7.3% after the reaction at 573 K and to 32.9% after the reaction at 593 K for 0.2 and 1.5 wt.% K-promoted catalysts, respectively. Shroff et al. [38], Huang et al. [39] and Bian et al. [40] have reported that Fe_3O_4 was the only phase in the used iron catalysts detectable by XRD analysis, because of the poor crystallographic form of iron carbides.

The MES spectra for catalysts with different potassium loadings after different reaction temperatures (shown in Fig. 8) have been fitted with four sextets, and the spectral parameters are summarized in Table 4. As mentioned above, the sextets with hyperfine field (Hhf) of 491–495 and 456–462 kOe can be attributed to tetrahedral site (A site) and octahedral site (B site) of magnetite, respectively. The fitted Hhf value of 209–223 kOe is consistent with that of Hägg carbide, $\chi\text{-Fe}_5\text{C}_2$ [41,42]; a new iron carbide phase, hexagonal close packing (HCP) carbide $\epsilon'\text{-Fe}_{2.2}\text{C}$ with the fitted Hhf value of 169–176 kOe, is found to have been formed during the reaction [10,42]. Although the ratios of S_B to S_A are still less than that of theoretical value of 2.0, the ratio of S_B to S_A in the catalysts after reaction is greatly larger than that in the reduced catalysts. The content of iron carbide gradually increases along with the process of FTS reaction, and it increases with the increase of both potassium content and reaction temperature. The results indicate that the increase in both potassium loading and temperature can improve the carbonization of the iron catalyst [43]. It implies that the catalysts are further reduced and carbonized during the reaction process.

3.3. Catalytic activity

The effect of potassium on the FTS activity at the reaction temperature range of 523–593 K, measured by carbon monoxide conversion, is shown in Fig. 9 and Table 5. The

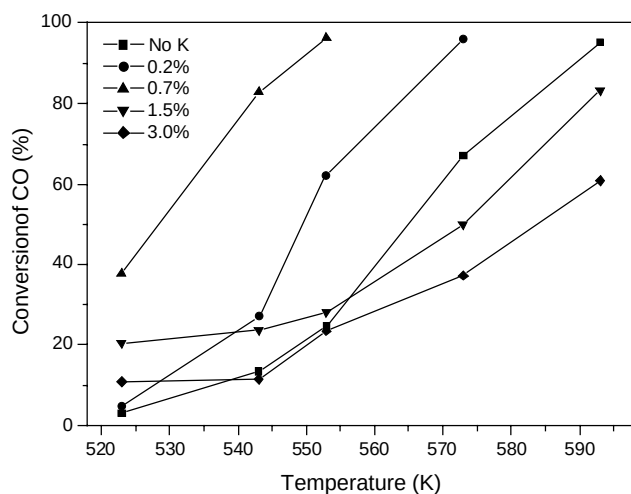


Fig. 9. Effects of potassium content and temperature on carbon monoxide conversion.

CO conversion increases significantly with the increase of potassium content and passes through a maximum at the potassium content of 0.7 wt.%. Beyond this potassium concentration, a monotonic decrease in catalyst activity is observed with the increase of potassium. CO conversion generally increases with the increase of reaction temperature. However, reaction temperature strongly improves the FTS activity over the lower potassium loading samples (weight percent of potassium less or equal to 0.7%) and un-promoted catalyst, while FTS activity is only slightly enlarged with the increase in reaction temperature for the catalysts with high potassium loadings. There is an increase of less than 5 and 8% in CO conversion for 1.5 and 3.0 wt.% potassium-promoted catalysts over the reaction temperature range of 523–553 K, respectively.

Numerous studies have been performed to investigate the effect of potassium on the FTS activity over various iron-based catalysts under different reaction conditions [3,4,7,12–14,18,20,21]. It was reported that FTS activity increases (or decreases) with the increase in potassium loading, passes through a maximum as a function of potassium content, or even has no direct relationship with potassium level. Bukur et al. [14] found that potassium significantly improved FTS activity within the range of 0–0.5K/100Fe, but beyond this concentration the effect of promotion was negligible. Anderson [3] found, however, that the activity of potassium promoted alumina supported iron catalyst was even lower than that of un-promoted catalysts. Pennline et al. [13] studied the promotion of potassium over a high Mn/Fe ratio (79Mn/21Fe) co-precipitated catalyst. They found that the 0.4 and 1.3 wt.% potassium-promoted catalysts had very high initial activity but decreased significantly with time on stream to values comparable to the un-promoted catalyst, and their activities were lower than that of un-promoted catalyst. The authors suggested that the potassium enhanced the deposition of carbon on the catalyst surfaces. Köbel [4] postulated that the reduced catalysts containing potassium had a higher concentration of active sites than that on un-promoted catalyst, and therefore accelerated the FTS activity. With further increase in the potassium content, the active sites may be blocked by potassium, resulting in a decline in catalyst activity. Furthermore, as stated earlier, the addition of potassium is in favor of carbon deposition on the surface, which leads to the formation of inactive carbon covering the active sites on the surface and thus leads to further decline in the FTS activity [7,20,44]. The effect of potassium on FTS activity observed in the present study is similar in some respects to that reported in some previous studies [12,21]. The investigation of Miller and Moskovits [21] indicated that there is a competition between dissociative CO chemisorption and H_2 adsorption on the active sites of catalysts, which results in a maximum in conversion as a function with the change in potassium content. At lower potassium levels, H_2 chemisorption is stronger than CO adsorption; thus the concentration of active C_1 species available for the formation of hydrocarbon is relatively lower, and the FTS activity is

Table 5
Effects of reaction temperature and potassium content on catalyst activity and selectivity^a

	K content (wt.%)															
	0		0.2				0.7				1.5				3.0	
Temperature (K)	550	573	593	543	553	573	543	553	543	553	573	543	553	573		
CO conversion (%)	21.9	66.9	95.3	27.1	62.3	95.9	87.8	96.2	23.6	28.1	51.1	14.5	23.4	37.2		
(H ₂ + CO) conversion (%) ^b	21.1	48.9	71.0	22.6	43.7	66.9	51.4	66.7	17.7	19.7	31.3	12.9	17.7	27.9		
Oxygenates (wt.% in total HC and oxygenates)	34.1	34.1	27.1	31.4	27.6	17.1	12.9	12.7	16.7	13.9	6.8	16.7	16.4	16.1		
CO ₂ selectivity (mol%)	35.9	38.8	33.4	35.6	36.5	40.1	40.4	36.4	22.8	33.2	40.8	20.6	21.9	28.8		
$P_{CO_2}P_{H_2}/P_{CO}P_{H_2O}$	1.9	4.0	28.4	2.1	4.7	38.1	22.3	50.9	2.7	3.1	6.1	1.9	2.3	2.7		
Hydrocarbon selectivities (wt.%)																
CH ₄	23.0	29.8	36.1	15.4	17.0	30.3	10.6	13.5	8.6	8.1	10.6	8.0	8.9	10.4		
C ₂ –C ₄	39.9	46.9	42.6	33.8	34.6	44.6	27.8	32.6	27.4	26.9	28.1	27.0	28.4	24.0		
C ₅ –C ₁₁	18.7	17.5	17.5	31.6	28.8	19.8	26.6	27.0	15.9	21.6	23.1	13.7	16.1	21.6		
C ₁₂ –C ₁₈	11.4	3.7	2.3	12.2	13.4	3.3	21.6	15.9	19.7	20.8	21.0	22.2	20.7	19.3		
C ₁₉ ⁺	7.0	2.2	1.5	7.0	6.2	2.0	13.5	11.1	28.4	22.6	17.2	29.1	25.9	24.7		
Olefin selectivity (wt.%) in C ₂ –C ₄																
C ₂	12.1	4.1	1.0	53.5	28.2	1.2	72.1	32.7	72.2	68.3	75.5	67.2	69.3	72.0		
C ₃	44.3	28.6	11.3	83.2	76.4	31.8	82.1	74.1	68.9	81.3	81.0	75.1	83.7	86.2		
C ₄	50.2	48.5	28.2	73.9	65.5	44.1	77.8	60.9	50.2	78.6	87.1	80.7	79.1	77.3		
C ₂ – ₄	31.7	22.2	10.8	70.3	55.7	22.9	78.2	57.0	59.3	75.2	79.3	75.4	77.2	79.0		

^a Reaction condition: 2.50 MPa, H₂/CO = 2.0, GHSV = 1000 h⁻¹.

^b (H₂ + CO) conversion defined as 100[amount (moles) of (H₂ + CO) converted]/[amount (moles) of (H₂ + CO)].

limited due to the lacking carbon sources on the surface. With the potassium increasing, CO dissociative adsorption is enhanced and the FTS activity increases. At moderate potassium levels, the CO chemisorption is enhanced, and at the same time, the H₂ chemisorption is not markedly suppressed, and thus an optimum activity is obtained. While the potassium level is greater than that required for optimum activity, the CO dissociative and associative adsorption are significantly increased, leading to the deposition of carbon on the surface, and the H₂ chemisorption is markedly suppressed, which resulted in a decline in FTS activity.

The present study also found that the increase in the reaction temperature slightly enlarges the FTS activities over the high potassium loading catalysts (1.5 and 3.0 wt.%) under the lower reaction temperature, which may be caused by the fact that the addition of potassium restrains the reduction of catalysts, as described in the characterization sections of XRD and MES. The catalysts are further reduced during FTS reaction. As illustrated by the XRD and MES analysis in the present study, the addition of potassium facilitates the formation of iron carbides during reaction, which is the active phase for FTS reaction. However, with further

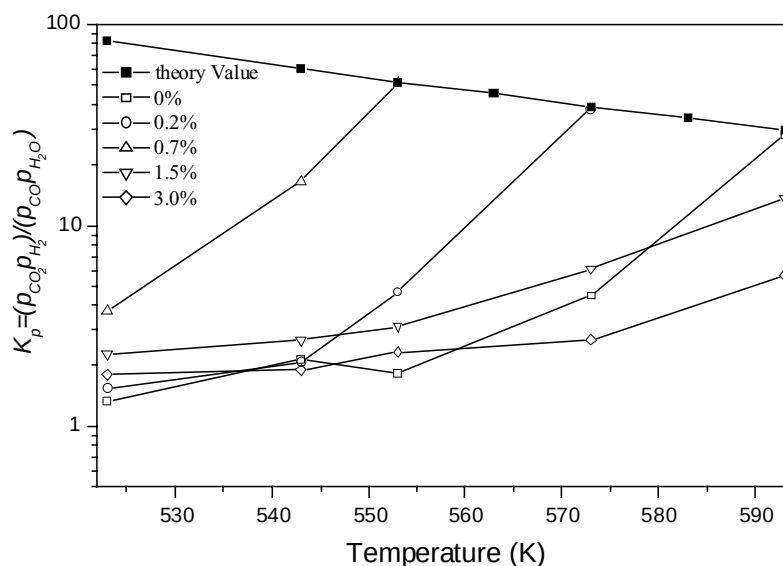


Fig. 10. Effects of potassium content and temperature on WGS activity.

increase in potassium loading, carbon deposition is significantly enhanced, and parts of the active sites are blocked on the catalyst surface.

A reversible water-gas shift (WGS) reaction accompanies the FTS reaction over iron catalyst. The WGS activity in the present study is represented by $K_p = P_{\text{CO}_2} P_{\text{H}_2} / P_{\text{CO}} P_{\text{H}_2\text{O}}$. The WGS activity values of catalysts with different potassium contents as the function of reaction temperature and the equilibrium value are shown in Fig. 10. The variation in the WGS activity with temperature and potassium level has a similar trend with that to FTS reaction activity. The mechanism by which potassium promotes WGS activity of iron catalyst is not well understood. In a general way, potassium could promote the WGS activity [7,14,18,20]. On the other hand, some researchers proposed that Fe_3O_4 is the active phase for WGS reaction [45]. The results of XRD and MES analysis in the present study indicate that the addition of potassium significantly accelerates the formation of carbides and further restrains the WGS reaction. As a synergistic effect of the results stated above, the WGS activity has a maximum as a function of potassium content in the catalyst.

3.4. Product selectivity

The effects of potassium content on the selectivity to olefins under different temperatures, represented as the weight ratios of $\text{C}_{2-4}^=$ to C_{2-4}^0 , are shown in Fig. 11 and Table 5. The primary effect of potassium promotion on the selectivity of olefin is to suppress the secondary hydrogenation of olefin [14]. The content of olefin increases with the increase of potassium loading and maintains a high olefin to paraffin ratio at higher potassium loading (K wt.% = 1.5 wt.%), as shown in Fig. 11 and Table 5. As temperature increases, the ratio of olefin to paraffin decreases over the un-promoted and lower potassium loading (weight percent of K = 0.7 wt.%), while the olefin to paraffin ratio is observed to decrease less or even to increase somewhat with increasing temperature over higher potassium loading catalysts (K wt.% = 1.5 wt.%).

The effect of potassium on the olefin selectivity in the present study is in general agreement with several earlier studies [14,18,20]. The influence of potassium on the selectivity of olefin is consistent with its effect on the strength of carbon monoxide and hydrogen chemisorptions. Addition of potassium increases the strength of CO chemisorption and suppresses that of H_2 . This results in a high concentration ratio of CO to H_2 on catalyst surface and consequently in a low hydrogenation activity and a high olefin selectivity of potassium-promoted catalysts [3,5,18,46]. As shown in Fig. 11, there is an obvious difference in the trend of olefin content for high potassium promoted and low potassium promoted catalysts under various reaction temperatures. The results from earlier studies [14,18,46] with iron catalysts are generally in agreement with those obtained in the present study. With the increase of reaction temperature,

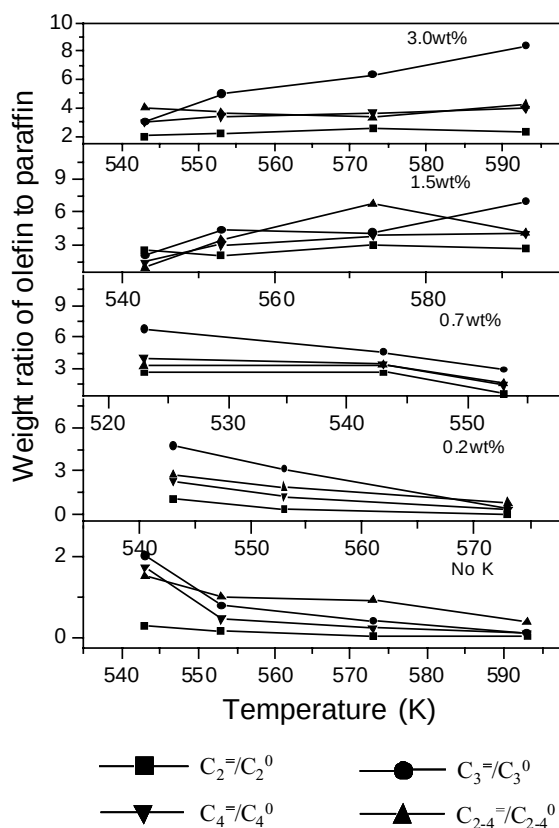


Fig. 11. Effects of potassium content and temperature on the olefin to paraffin ratio.

the rates of both primary (olefin formation) and secondary hydrogenation of olefins (secondary reaction) are expected to increase. Consequently, the selectivity may increase, decrease, or even maintain a constant value depending on the relative rates of 1-olefin formation (primary reaction) and olefin hydrogenation (secondary reaction). The high loading potassium enhances CO chemisorption and restrains H_2 chemisorption. This results in a high rate of olefin formation and low hydrogenation activity. Therefore, the selectivity of olefins will increase with reaction temperature for high potassium loading catalysts.

The selectivity of oxygenates, mainly composed of alcohols and esters represented as weight percent of oxygenate in the total hydrocarbon and oxygenate, is summarized in Table 5. The selectivity of oxygenates in the products shows a monotonic decrease with the increase of potassium loading and passes through a minimum at 0.7 wt.%. After the minimum point, the selectivity increases slowly with the further increase in potassium content. At the same time, increasing reaction temperature results in a monotonic decrease in the weight percent of oxygenates over the un-promoted and potassium-promoted catalysts. This fact is in agreement with the assumption of Bileon and Sachtler [47] that oxygenates are formed on oxidic patches of the catalysts surface. By comparing with MES and XRD spectra of the used catalysts in present study, one finds that the selectivity of oxygenates

varies in proportion to the content of iron oxide and is inversely proportional to that of iron carbides. The result of characterization agrees well with the explanation of Bileon and Sachtler.

3.5. Hydrocarbon distribution

Table 5 shows the effect of potassium on the distribution of hydrocarbons under different reaction temperatures. Table 5 indicates that the average molecular weight of FTS products increases with the decrease of temperature. Potassium is thus an effective promoter to restrain the formation of methane and gaseous products and to shift selectivity to higher molecular weight hydrocarbons. The effect of potassium on the hydrocarbon distribution observed in the present study is in good agreement with the results obtained in several earlier studies with a variety of iron-based FTS catalysts [3,5,14,18,22,48]. The increase in average molecular weight of hydrocarbon products is due to the fact that the CO/H concentration ratio on catalyst surface can be increased by potassium addition. Therefore, the presence of potassium enhances the probability of continued chain growth, and forms higher molecular weight hydrocarbons [4,8,14]. The present study also found that this restraining effect of potassium on the formation of methane and gaseous products is significant. In several earlier studies of Kölbel [4] and Bukur et al. [14], they also observed an increase in formation of higher molecular hydrocarbons over potassium-promoted catalysts. In our study, there is no obvious change in the selectivity to methane and gaseous hydrocarbons between the catalysts with the higher potassium content (1.5 and 3.0 wt.%).

4. Conclusions

Promotion of Fe/Mn catalyst with potassium in the range of 0–3.0 wt.% causes an increase in the crystallite size of catalyst and a decrease in BET surface area. With the increase of potassium content, the reduction of the Fe/Mn catalyst is retarded due to the strong interaction of iron oxide with potassium oxide, and catalysts after reduction only contain small amount of iron carbides (χ -Fe₅C₂). The process of FTS reaction, however, accelerates the carbonization of iron catalysts and ϵ' -Fe_{2.2}C is also formed during the run of FTS. A maximum in catalytic activity (FTS and WGS) is obtained at a particular level of potassium (0.70 g of K per 100 g); there is a decline in activity at potassium levels in excess of the optimum.

Potassium is an effective promoter to restrain the formation of methane and gaseous products, and to shift selectivity to higher molecular weight hydrocarbons. It also suppresses olefin hydrogenation, which leads to an increase in olefin content in the products. The selectivity to oxygenates in the products shows a rapid monotonic decrease with the increase of potassium loading and passes through a minimum at 0.7 wt.% of potassium loading. After the minimum point,

it increases slowly with further increasing of potassium content. At the same time, increasing reaction temperature results in a monotonic decrease in the weight percent of oxygenates over the un-promoted and potassium-promoted catalysts.

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